Forensic Biometrics & Master Persona Directory

ALL IDENTITIES CLASSIFIED AS UNKNOWN SUBJECTS (PENDING ENROLLMENT)
Consolidated Ground-Truth Registry: **43 Real Unique Individuals** (REAL-01 to REAL-43). DBSCAN clustering originally generated 1,652 micro-clusters due to off-angle CCTV variations. Batched ArcFace pairwise biometrics and connected graph analysis mathematically consolidated these into 43 true human beings. All individuals are cataloged below.

1. Forensic Biometric Resolution Statement

Biometric feature analysis confirmed that the unknown individual observed operating the multiport backwash valve on Day 1 at **09:53:38** (originally missing from automated motion ingest and enrolled as SUBJ-1652) shares high cosine similarity (>= 0.55 - 0.69) with SUBJ-10, SUBJ-1079, SUBJ-759, and SUBJ-582, proving that they are the same recurring maintenance individual (**REAL-01**).
Furthermore, ArcFace cosine similarity between this individual and **REAL-02** (anchor SUBJ-02) is **0.1831** (non-match), establishing that they are entirely distinct persons and resolving all previous ambiguity.

2. Consolidated Real Persons Master Directory (All 43 Individuals)

The directory below catalogs all **43 real unique individuals** identified across the 30-day surveillance period:

Real ID	Anchor	Frags	Dets	Days	Contacts	Classification	Risk
REAL-01	SUBJ-10	19	505	18	**7**	Core Recurring	**CRITICAL**
REAL-02	SUBJ-02	2	168	13	**54**	Core Recurring	**CRITICAL**
REAL-03	SUBJ-16	2	67	8	**0**	Core Recurring	**LOW**
REAL-04	SUBJ-342	2	6	1	**4**	Transient	**ELEVATED**
REAL-05	SUBJ-01	1	29,598	29	**116**	Core Recurring	**CRITICAL**
REAL-06	SUBJ-03	1	104	2	**0**	Core Recurring	**LOW**
REAL-07	SUBJ-04	1	87	16	**0**	Core Recurring	**LOW**
REAL-08	SUBJ-05	1	86	2	**0**	Core Recurring	**LOW**
REAL-09	SUBJ-06	1	85	12	**0**	Core Recurring	**LOW**
REAL-10	SUBJ-07	1	81	3	**0**	Core Recurring	**LOW**
REAL-11	SUBJ-08	1	79	10	**0**	Core Recurring	**LOW**
REAL-12	SUBJ-09	1	79	1	**0**	Core Recurring	**LOW**
REAL-13	SUBJ-11	1	69	9	**0**	Core Recurring	**LOW**
REAL-14	SUBJ-12	1	68	2	**0**	Core Recurring	**LOW**
REAL-15	SUBJ-13	1	47	5	**0**	Core Recurring	**LOW**
REAL-16	SUBJ-18	1	41	2	**0**	Periodic	**LOW**
REAL-17	SUBJ-19	1	39	2	**0**	Periodic	**LOW**
REAL-18	SUBJ-22	1	35	1	**1**	Periodic	**ELEVATED**
REAL-19	SUBJ-23	1	33	2	**0**	Periodic	**LOW**
REAL-20	SUBJ-24	1	32	1	**0**	Periodic	**LOW**
REAL-21	SUBJ-25	1	29	9	**0**	Core Recurring	**LOW**
REAL-22	SUBJ-28	1	27	3	**0**	Periodic	**LOW**
REAL-23	SUBJ-29	1	27	1	**0**	Periodic	**LOW**
REAL-24	SUBJ-31	1	26	1	**0**	Periodic	**LOW**
REAL-25	SUBJ-33	1	23	6	**0**	Core Recurring	**LOW**
REAL-26	SUBJ-34	1	23	1	**0**	Periodic	**LOW**
REAL-27	SUBJ-36	1	22	1	**0**	Periodic	**LOW**
REAL-28	SUBJ-37	1	22	1	**0**	Periodic	**LOW**
REAL-29	SUBJ-38	1	21	10	**0**	Core Recurring	**LOW**
REAL-30	SUBJ-39	1	21	1	**0**	Periodic	**LOW**
REAL-31	SUBJ-40	1	21	1	**0**	Periodic	**LOW**
REAL-32	SUBJ-41	1	20	10	**4**	Core Recurring	**ELEVATED**
REAL-33	SUBJ-42	1	20	1	**0**	Periodic	**LOW**
REAL-34	SUBJ-43	1	20	1	**0**	Periodic	**LOW**
REAL-35	SUBJ-46	1	20	1	**0**	Periodic	**LOW**

Real ID	Anchor	Frags	Dets	Days	Contacts	Classification	Risk
REAL-36	SUBJ-47	1	19	2	**0**	Periodic	**LOW**
REAL-37	SUBJ-49	1	19	1	**0**	Periodic	**LOW**
REAL-38	SUBJ-102	1	8	1	**5**	Transient	**CRITICAL**
REAL-39	SUBJ-129	1	7	1	**2**	Transient	**ELEVATED**
REAL-40	SUBJ-167	1	6	1	**5**	Transient	**CRITICAL**
REAL-41	SUBJ-489	1	2	1	**2**	Transient	**ELEVATED**
REAL-42	SUBJ-890	1	2	1	**1**	Transient	**ELEVATED**
REAL-43	SUBJ-993	1	2	1	**1**	Transient	**ELEVATED**

3. Critical Equipment Protection Zone Breach Ledger (All 18 Unknown Contact Breaches)

Across the 30-day archive, exactly **18 unique unknown individuals** established verified direct physical contact (IoU > 0.08) with plant equipment assets:

Subject ID	Contacts	Active Days	Total Dets	First Seen	Last Seen	Primary Asset Contacted	Hazard
SUBJ-01	**116**	29	29598	2026-09-01	2026-09-30	Two Multiport Backwash Valves	**CRITICAL**
SUBJ-02	**53**	12	166	2026-09-01	2026-09-30	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-102	**5**	1	8	2026-09-02	2026-09-02	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-167	**5**	1	6	2026-09-25	2026-09-25	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-41	**4**	10	20	2026-09-03	2026-09-24	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-344	**3**	1	3	2026-09-09	2026-09-09	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-395	**3**	1	3	2026-09-16	2026-09-16	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-129	**2**	1	7	2026-09-06	2026-09-06	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-489	**2**	1	2	2026-09-01	2026-09-01	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-22	**1**	1	35	2026-09-19	2026-09-19	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-342	**1**	1	3	2026-09-09	2026-09-09	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-582	**1**	1	2	2026-09-02	2026-09-02	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-759	**1**	1	2	2026-09-09	2026-09-09	Ultimate Plant Power Switch	**SEVERE**
SUBJ-842	**1**	1	2	2026-09-13	2026-09-13	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-890	**1**	1	2	2026-09-15	2026-09-15	Two Multiport Backwash Valves	**CRITICAL**
SUBJ-993	**1**	1	2	2026-09-19	2026-09-19	RO Skid Controls & Walkway	**ELEVATED**
SUBJ-1079	**1**	1	2	2026-09-24	2026-09-24	Ultimate Plant Power Switch	**SEVERE**
SUBJ-1652	**1**	1	28	2026-09-01	2026-09-01	Two Multiport Backwash Valves	**CRITICAL**